Clinical قصد ٥ج Clinical pharmacy 2025

10-1

Acute Diarrhea

Normal bowel frequency ranges from three times a day to three times a week in the normal population.

Acute diarrhea is defined as a greater number of stool frequency or increased fluidity from the normal lasting for less than 14 days.

If the illness persists for more than 14 days, it is called persistent diarrhea.

If the duration of symptoms is longer than 1 month, it is considered chronic diarrhea.

Prevalence:

Worldwide, acute diarrhea constitutes a major cause of morbidity and mortality, especially among the very young, very old. Most cases are caused by enteric infections.

Pathophysiology:

Approximately 8 to 9 L of fluid enter the intestines daily--1 to 2 L represents food and liquid intake, and the rest is from endogenous sources such as salivary, gastric, pancreatic, biliary, and intestinal secretions.

Most of the fluid, about 6 to 7 L, is absorbed in the small intestine, and only about 1 to 2 L is presented to the colon. Most of this is absorbed as it passes through the colon, leaving a stool output of about 100 to 300 g/day.

Although many organisms simply impair the normal absorptive processes in the small intestine and colon, others, organisms, such as Vibrio cholerae, secrete a toxin that causes the colonic mucosa to secrete, rather than absorb, fluid and electrolytes. Voluminous diarrhea may result.

Causes of Acute Diarrhea:

i. Infections:

1- Viral Infections

- Many food-borne viruses e.g. Rotavirus, which is the most common cause of diarrhea in children& Adenovirus & Norwalk virus.
- Cytomegalovirus & Herpes simplex virus.

2- Bacterial Infections: Several types of bacteria can cause acute diarrhea. These include:

- Common causes : Shigella & Campylobacter jejuni & Escherichia coli (E. coli)
- Vibrio cholerae.
- Clostridium perfringens.
- Food poisoning: Staphylococcus aureus (S. aureus) & Bacillus cereus& Salmonella species.

3- Parasitic infection: Parasites can enter the body through food or water:

- Giardia lamblia.
- Entamoeba histolytica.
- Cryptosporidium.

ii. Medications: Some of these medicines include:

- Antibiotics, laxatives, antacids containing magnesium.
- Other substances that may cause this condition include: Alcohol, Caffeine, Herbal teas.

iii. Functional Conditions

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- The functional condition that most often results in acute diarrhea is anxiety and stress (Irritable bowel syndrome).

Common food sources of bacterial infection (food poising):

- Shigellosis: Common in daycare settings.
- V. Cholera: common in some developing countries, mainly by water.
- Salmonella species: Contaminant of eggs and poultry (S. typhimurium).
- Staph aureus: Common contaminant of processed meat.
- Bacillus cereus: Contaminant of rice and bean.
- Cambylobacter jejuni: Contaminant of poultry and in household pets.
- E. coli: Undercooked or raw hamburger.
- C. perferingens: Common in food that have been warmed on steam table.
- Yersinia eneterolitica: By eating infected meat or dairy products.
- Vibrio parahemolyticus: Eating raw seafood especially oysters.

Symptoms and signs

Clinical features sometimes provide a clue to the cause. Diarrhea caused by small intestine disease is typically high volume, watery, and often associated with malabsorption. Dehydration is frequent. Diarrhea caused by colonic involvement is more often associated with frequent small-volume stools, the presence of blood, and a sensation of urgency.

Important factors in evaluating acute diarrhea:

Travel history, sources of water (e.g., well water), recent food intake, symptom duration, including bowel movement frequency, stool quantities, dysentery with fever, tenesmus, hematochezia or pus in the stool, signs of volume depletion, including thirst, tachycardia, orthostatic hypotension, decreased urine output, skin turgor, and lethargy or confusion, or both.

Patients ingesting toxins or those with toxigenic infection typically have nausea and vomiting as prominent symptoms, along with watery diarrhea, but rarely have a high fever.

Clues for certain infections:

- Vomiting that begins within 6 hours of ingesting a food should suggest food poisoning caused by preformed toxin from bacteria such as S. aureus or B. cereus.
- If diarrhea disease begins within 8 to 14 hours of food ingestion, C. perfringens should be suspected.
- When the incubation period is longer than 14 hours and vomiting is also a significant symptom, along with the diarrhea, viral agents should be considered.
- Parasites that do not invade the intestinal mucosa, such as Giardia lamblia and Cryptosporidium, usually cause only mild abdominal discomfort. Giardiasis may be associated with mild steatorrhea, gaseousness, and bloating.
- Infection with invasive bacteria such as Campylobacter, Salmonella, and Shigella spp., and organisms that produce cytotoxins, such as C. difficile and enterohaemorrhagic E.

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coli (serotype O157:H7), frequently result in abdominal pain, and low-grade fever; occasionally, peritoneal signs may suggest a surgical abdomen.

- Yersinia organisms often infect the terminal ileum and caecum and manifest with right lower quadrant pain and tenderness, suggestive of acute appendicitis.

Diagnosis & Investigations:

Acute diarrheas are usually infectious in origin and, for the most part, resolve with or without intervention before a diagnosis is made.

-The presence of blood is a useful clue, suggesting infection by invasive organisms, inflammation, ischemia, or neoplasm.

-Large-volume diarrhea suggests small bowel or proximal colonic disease, whereas small frequent stools associated with urgency suggest left colon or rectal disease. All current and recent medications should be reviewed, specifically new medications, antibiotics, antacids, and alcohol abuse.

A medical evaluation of acute diarrhea is not warranted in the previously healthy individual if symptoms are mild, moderate, spontaneously improve within 48 hours, and are not accompanied by fever, chills, severe abdominal pain, or blood in the stool

Indications to investigate a case of Acute Diarrhea:

- Evaluation is indicated if symptoms are severe or prolonged, the patient appears "toxic," there is evidence of colitis (occult or gross blood in the stools, severe abdominal pain or tenderness, and fever), or empirical therapy has failed.
- Passage of many small-volume stools containing blood and mucus, temperature higher than 38.5° C, passage of more than six unformed stools in 24 hours, or a duration of illness longer than 48 hours, diarrhea with severe abdominal pain in a patient older than 50 years, diarrhea in older adults (older than 70 years) .
- The physical examination Vital signs (including temperature and orthostatic evaluation of pulse and blood pressure) and signs of volume depletion (including dry mucous membranes, decreased skin turgor, and confusion) should be carefully evaluated.
- A careful abdominal examination to evaluate tenderness and distention.

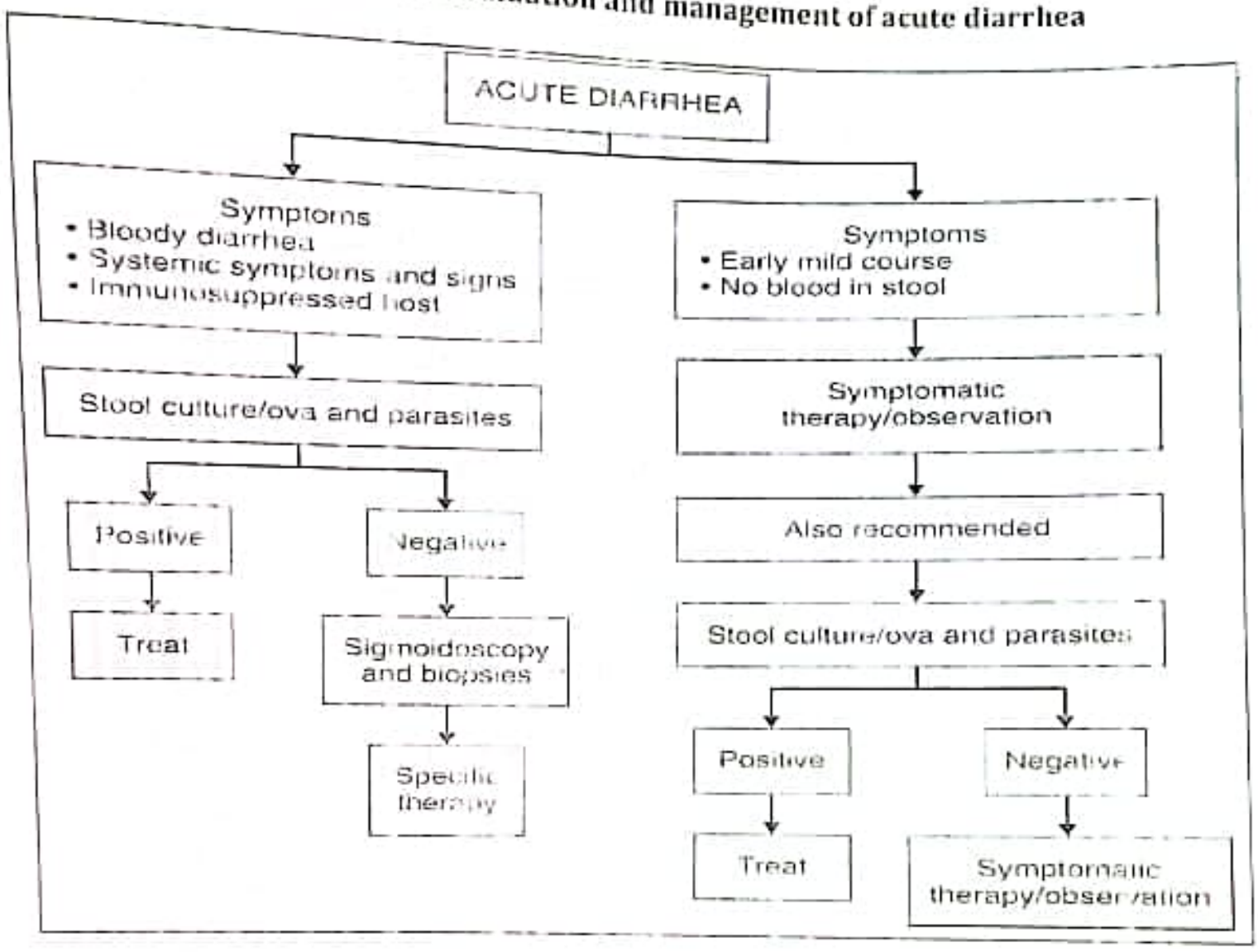
Investigation of a case of Acute Diarrhea:

- Stool analysis: Fecal leukocyte determination, Stool culture for enteric pathogens, Stool examination for ova and parasites. Stool evaluation for fecal leukocytes (or lactoferrin, a by-product of white blood cells) is a useful initial test, because it may support a diagnosis of inflammatory diarrhea. If the test is negative, stool culture may not be necessary, but culture is indicated if the test is positive.

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- Stool testing for ova and parasites Multiple fresh stool collections should be collected at different times because shedding of parasites may be intermittent.
- Flexible sigmoidoscopy with biopsy (sometimes it may be contraindicated).

Algorithm for the evaluation and management of acute diarrhea



Treatment:

1- Resuscitation and supportive therapy

- Fluid and electrolyte replacement, dietary modifications (Rehydration).
- Diet: Total food abstinence is unnecessary and not recommended. Foods providing calories are necessary to facilitate renewal of enterocytes.
- Diet should be avoided: Dairy products should be avoided: transient lactase deficiency can be caused by viral and bacterial infections. Caffeinated beverages and alcohol should be avoided as it enhance intestinal motility and secretions.

2- Pharmacologic Measures:

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A- **Antidiarrheal Agents: Antimotility:** These can be useful for the amelioration of symptoms. The most effective agents are the opioid derivatives e.g. Loperamide, Diphenoxylate and tincture of opium.

- These agents should be avoided in patients with fever, bloody diarrhea, and possible inflammatory diarrhea (prolonged fever in patients with shigellosis, toxic megacolon in patients with C. difficile infection).
- Bismuth subsalicylate (bulk forming). somewhat less effective than loperamide. It is usually used in traveler's diarrhea.

B- **Antimicrobial Treatment:** Because most patients have mild. self-limited disease caused by viruses or noninvasive bacteria. routine empirical treatment is not warranted.

Empirical antibiotic treatment is indicated in:

- Patients with suspected invasive bacterial infection. traveler's diarrhea. or immunosuppression and poor general condition.
- C. difficile–associated diarrhea.
- Febrile patient with fecal leukocytes.
- Giardia can be prescribed for those with a 2-week or longer history of diarrhea.
- Moderate or severe diarrhea.
- Hospitalization e.g. dehydration or with Symptoms continue for > one week.

Example for specific antibiotics:

- For traveler's diarrhea: Rifaximin.
- Shigella or salmonella species: Quinolone.
- Enteropathogenic E. coli: Quinolone.
- Campylobacter: Erythromycin.
- Intestinal E. histolytica: Metronidazole.
- Giardia lamblia: Metronidazole.

C- **Probiotics:** which are defined as live microorganisms that, when administered in adequate amounts, confer a health benefit on the host. They are given oral as capsules. The commercially available strains are from the Bifidobacterium and Lactobacillus.

Prevention of diarrhea:

- Food safety: follow good hygienic measures e.g wash hands and knives.
- Keep the refrigerator temperature at 40°F (4.4°C) or lower.
- Cook raw food from animal sources to a safe internal temperature.
- When eating always remember: Boil it, cook it, peal it or forget it.

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Chronic Diarrhea

Diarrhea persistent for more than four weeks is considered chronic diarrhea.

Pathophysiology of Diarrhea:

I) Osmotic Diarrhea.

- Osmotic diarrhea is caused by the accumulation of non absorbed solutes in the gut lumen.
- Osmotic diarrhea stops when the patient fasts.
- The fecal fluid has a **large solute gap**, i.e., (Low electrolytes in stool).

Causes:

1- Ingestion of poorly absorbable solutes: Magnesium sulfate laxative. Mannitol, sorbitol.
2- Maldigestion. Disaccharidase deficiencies, short bowel syndrome, postgastrectomy, postvagotomy state, and chronic intestinal ischemia.

(II) Secretory Diarrhea:

- Stools are large in volume and watery and diarrhea persists with fasting. High electrolytes in stool (**small solute gap**). Due to active substance act as secretory inducer.

Causes:

- VIP (Vaso active Intestinal Peptides) production usually comes from tumors originating in pancreatic islet cells, VIPoma syndrome. Carcinoid syndrome, Serotonin secreting tumors, Medullary carcinoma of the thyroid, Zollinger-Ellison syndrome: gastrin.
- Bile acid diarrhea & ileal disease or resection.

(III) Deranged Intestinal Motility:

- Abnormally reduced peristalsis may allow bacterial overgrowth in the small bowel.
- "Intestinal hurry".
- Premature emptying of the colon: irritable bowel syndrome, diabetic neuropathy, colonic tumor, fecal incontinence.
- The most frequent causes for sphincter incontinence are previous anal surgery for fissures, fistulas, or hemorrhoids; episiotomy or tear during childbirth; anal Crohn's disease.

(IV) Diarrhea Due to Disruption of Structural Integrity of the Intestine

- Infectious: Sprue (tropical, nontropical) & Whipple's disease& Acquired immunodeficiency syndrome. Giardia, Cryptosporidium & Ileocecal tuberculosis & Diverticulosis.
- Non-infectious: Radiation enteritis & Amyloidosis & Collagen vascular diseases &. Inflammatory bowel disease &. Eosinophilic gastroenteritis & Intestinal lymphoma & Intestinal ischemia, mesenteric vasculitis.

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(V) - Miscellaneous Causes of Diarrhea:

- Diuretics, cardiac glycosides, propranolol, colchicine, methotrexate, 6-mercaptopurine, 5-fluorouracil, guanethidine, ethanol.
- Endocrine: Addison's disease, hypoparathyroidism.
- Neurologic: Tabes dorsalis, multiple sclerosis, myelitis, encephalitis, heat stroke, myotonia dystrophica, orthostatic hypotension.
- Toxins: Lead poisoning.
- Microscopic Colitis.

Some patients with chronic diarrhea demonstrate inflammation of colonic mucosa despite a normal appearance of the colon on barium enema and colonoscopy, but endoscopic biopsy may reveal abnormal lymphocytes infiltrations.

Diagnosis:

- **Clinical Evaluation**
- When the stools are consistently large in volume, the underlying cause of diarrhea is likely to be located in the small bowel or in the proximal colon. By contrast, small-volume diarrhea, in which the patient has frequent urges to defecate but passes only small amounts of feces or mucus, the disorder is usually in the left portion of the colon and rectum.
- Passage of blood mixed with the diarrheal stool usually indicates inflammation of the mucosa, less often a neoplasm.
- Passage of non bloody mucus suggests irritable bowel syndrome, as does a history of small-volume diarrhea alternating with constipation. Frothy stools and excessive flatus suggest fermentation of unabsorbed carbohydrates.
- Excessively foul stools suggest putrefaction of unabsorbed amino acids and amoebiasis
- Visible oil or fat indicates severe steatorrhoea.
- Fecal soiling (incontinence) suggests an anal sphincter defect.
- Diarrhea in a patient with features of anorexia nervosa suggests laxative abuse.

There are, of course, many other pertinent facts obtainable from the history, including previous surgery, drug intake, symptoms of systemic illness, travel, and related illnesses in family members. In chronic and recurrent diarrhea, an association of exacerbation of diarrhea with emotional stress should be sought, an association that may suggest irritable bowel syndrome.

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Diagnostic Tests:

1- Routine Examination of Stool.

- Microscopic examination: pus, parasite.
- Occult Blood: Occult (or gross) blood in association with diarrhea usually indicates inflammation.
- Stain for Fat (Sudan stain).
- Alkalinization: A pink color following alkalinization of a stool or urine sample indicates phenolphthalein ingestion.

2- Other Tests (specific according to the suspected etiology):

- Biochemical tests: serum glucose,T3,T4 ,serum amylase.
- Determination of urinary 5-hydroxyindole acetic acid....Carcinoid.
- Parasitic Organisms. Duodenal or jejunal biopsy or aspirate for Giardia, cryptosporidium.
- Proctosigmoidoscopy is helpful in establishing the presence or absence of mucosal inflammation.
- Rectal Biopsy. Amyloidosis. Whipple's disease, microscopic colitis.
- Quantitative Fecal Fat. Collected stools (usually for 72 hours)
- Vasoactive Intestinal Polypeptide (VIP) and other Secretory hormones in blood.

3- Therapeutic Trials:

These trials may include pancreatic enzymes, antibiotics, Metronidazole or anti-T.B drugs, cholestyramine, indomethacin (for prostaglandin synthetase inhibition), and various diets regimen to combat food allergy (lactose free, carbohydrate free, low fat, gluten free).

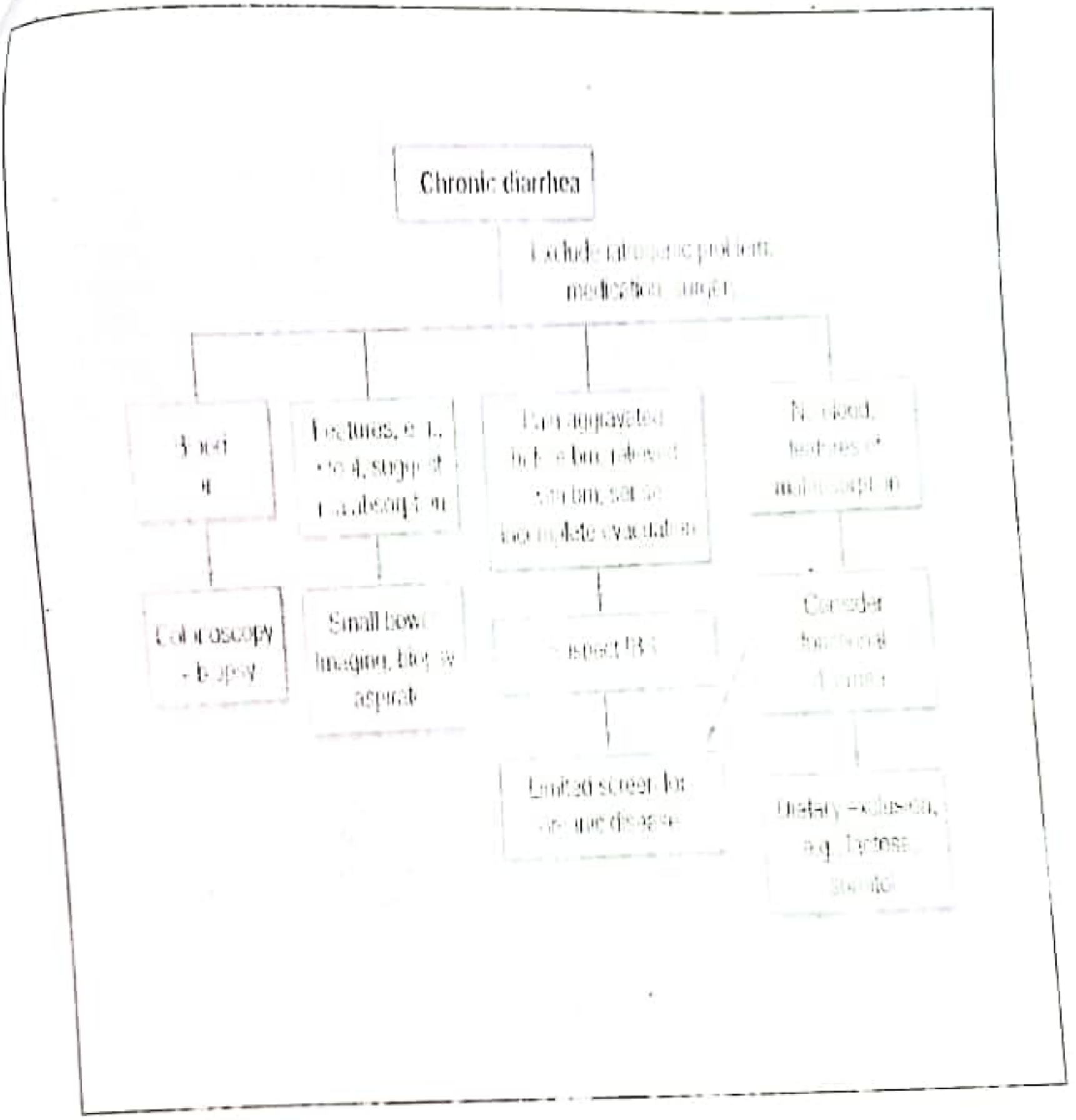
Therapy:

- The most satisfactory therapy is to cure the underlying disease.
- Supportive and symptomatic therapy: e.g. correction of vitamin deficiency...etc
- Opiates:

✓ Indicated in unresponsive –noninfectious diarrhea (e.g. autonomic neuropathy).
✓ Drugs: e.g. codeine, diphenoxylate with atropine (Lomotil), and loperamide.

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Algorithmic for approach of chronic diarrhea



IBS: Irritable bowel syndrome,
bm: bowel motion
Pr: per rectal

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